

CHE 201 Fall 2019 HW 9

Dennis Holm

TOTAL POINTS

100 / 100

QUESTION 1

11 15 / 15

- ✓ - **0 pts** Correct
- **15 pts** no submission
- **6 pts** v2 calculation incorrect
- **6 pts** P2 calculation incorrect

QUESTION 2

22 15 / 15

- ✓ - **0 pts** Correct
- **15 pts** Missing answer
- **1 pts** Wrong calculation. $W = -25.8 \text{ kW}$
- **3 pts** $h_1 = 417.46 \text{ kJ/kg}$. $W = -25.8 \text{ kW}$
- **6 pts** $(z_1 - z_2) = -25 \text{ m}$ (negative). $W = -25.8 \text{ kW}$
- **3 pts** $h_3 = 3230.9 \text{ kJ/kg}$. Mass flow rate of steam = 0.0616 kg/s

QUESTION 3

33 20 / 20

- **0 pts** Correct
- ✓ - **5 pts** power requirement = $-W$
- **20 pts** No submission
- **8 pts** wrong calculation for W
- ✓ + **5 pts** Extra points from Question 6

QUESTION 4

44 15 / 15

- ✓ - **0 pts** Correct
- **15 pts** Missing answer
- **2 pts** Did not define the control volumes
- **2 pts** Did not state the engineering model
- **3 pts** Should have used m_2 to calculate work. $W = 199.9 \text{ hp}$
- **3 pts** $h_1 \sim h_f(180 \text{ oC}) = 763.22 \text{ kJ/kg}$. $m_2 = 0.371 \text{ kg/s}$

- **1 pts** $m_2 = ?$
- **1 pts** Work is positive

QUESTION 5

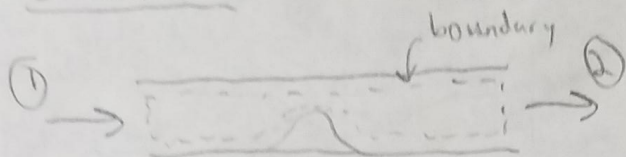
55 15 / 15

- **0 pts** Correct
- **15 pts** Missing or wrong answer
- ✓ - **2 pts** $W_{1-2} = 0.002 \text{ kJ}$
- **7.5 pts** b) $t = 1328.3 \text{ s}$ (22.1 min)
- ✓ - **2 pts** $W_{2-3} = -0.0522 \text{ kJ}$
- ✓ - **2 pts** $h_e = 2675.5 \text{ kJ/kg}$
- **3 pts** b) calculation is not clear.
- **13 pts** Incomplete answer
- **2 pts** $u_2 = 417.36 \text{ kJ/kg}$
- ✓ + **6 pts** Extra points from Question 6

QUESTION 6

66 20 / 20

- **0 pts** Correct
- **20 pts** Missing answer
- ✓ + **20 pts** Extra points for making the board
- **10 pts** Missing or incomplete explanation of how it works and/or missing schematics of important parts (eg: nozzle, fan, switch, heating elements, cord etc).
- **5 pts** Missing source for the drawing
- **2 pts** Missing schematics of the air flow
- **10 pts** Missing drawing
- Wow! Great job!

1.)
(4.65)Schematic:Engineering Model:

Control Volume
Steady State
Ideal Air
1 inlet / 1 outlet
 $\Delta PE = 0$

$$\dot{m} = \dot{Q} = 0$$

$$\frac{dm_{cv}}{dt} = 0 = \dot{m}_i - \dot{m}_e$$

$$\dot{m}_i = \dot{m}_e = \dot{m}$$

$$\frac{dE_{cv}}{dt} = 0 = \dot{m} \left(h_1 - h_2 + \frac{V_1^2 - V_2^2}{2} \right)$$

$$\Rightarrow 0 = h_1 - h_2 + \frac{V_1^2 - V_2^2}{2}$$

$$0 = 320.29 \frac{\text{kJ}}{\text{kg}} - 305.22 \frac{\text{kJ}}{\text{kg}} + \frac{(30 \frac{\text{m}}{\text{s}})^2 - V_2^2}{2} \cdot \frac{1 \text{ kg}}{1 \text{ kg}} \cdot \frac{1 \text{ kJ}}{1000 \text{ J}}$$

$$\boxed{V_2 = 176.2 \text{ m/s}}$$

$$b) \quad \dot{m} = \frac{A \cdot V}{N} \quad pN = \frac{R \cdot T}{M}$$

$$\dot{m} = \frac{A \cdot V \cdot M \cdot p}{R \cdot T}$$

$$\frac{A_1 \cdot V_1 \cdot M \cdot p_1}{R \cdot T_1} = \frac{A_2 \cdot V_2 \cdot M \cdot p_2}{R \cdot T_2}$$

$$p_2 = \frac{V_1 \cdot p_1 \cdot T_2}{T_1 \cdot V_2}$$

$$\boxed{p_2 = 160.8 \text{ kPa}}$$

Given:

$$T_1 = 320 \text{ K}$$

$$p_1 = 900 \text{ kPa}$$

$$V_1 = 30 \text{ m/s}$$

$$T_2 = 305 \text{ K}$$

From Table A-22:

$$h_1 = 320.29 \text{ kJ/kg}$$

$$h_2 = 305.22 \text{ kJ/kg}$$

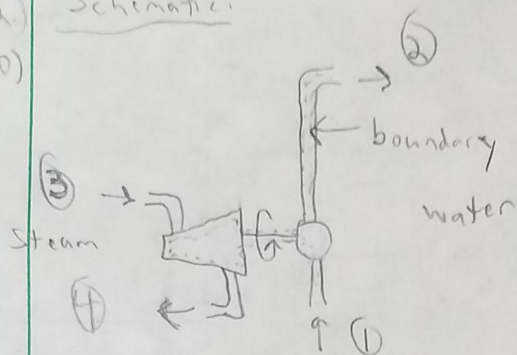
11 15 / 15

✓ - 0 pts Correct

- 15 pts no submission

- 6 pts v2 calculation incorrect

- 6 pts P2 calculation incorrect

2)
(4.70)Schematic:Eng. Model:

2 Control Volumes

Steady state

1 inlet / 1 outlet each

shaft work

$$\Delta PE_{34} = \Delta KE_{34} = 0$$

$$\dot{Q}_{34} = 0$$

$$\Delta KE_{12} = 0$$

$$\frac{dm_{12}}{dt} = 0 = \dot{m}_1 - \dot{m}_2$$

$$\dot{m}_1 = \dot{m}_2$$

$$\frac{dm_{34}}{dt} = 0 = \dot{m}_3 - \dot{m}_4$$

$$\dot{m}_3 = \dot{m}_4$$

$$\frac{dE_{12}}{dt} = \dot{Q}_{12} - \dot{W}_{12} + \dot{m}_1(h_1 - h_2 + g(z_1 - z_2)) = 0$$

$$\frac{dE_{34}}{dt} = -\dot{W}_{34} + \dot{m}_3(h_3 - h_4) = 0$$

$$\dot{W}_{12} = -\dot{W}_{34}$$

$$a.) \quad \dot{W}_{12} = \dot{Q}_{12} + \dot{m}_1(h_1 - h_2 + g(z_1 - z_2))$$

$$\dot{W}_{12} = -2 \text{ kW} + 50 \frac{\text{kg}}{\text{s}} \left(417.46 \frac{\text{kJ}}{\text{kg}} - 417.69 \frac{\text{kJ}}{\text{kg}} + 9.81 \frac{\text{m}}{\text{s}^2} (0 \text{ m} - 25 \text{ m}) \right) \dots$$

$$\dot{W}_{12} = -25.76 \text{ kW}$$

$$b.) \quad \dot{W}_{34} = \dot{m}_3(h_3 - h_4)$$

$$-\dot{W}_{12} = \dot{m}_3(h_3 - h_4)$$

$$25.76 \text{ kW} = \dot{m}_3 \left(3230.9 \frac{\text{kJ}}{\text{kg}} - 2812.0 \frac{\text{kJ}}{\text{kg}} \right)$$

$$\dot{m}_3 = 0.0615 \frac{\text{kg}}{\text{s}}$$

Given:

$$P_1 = 1 \text{ bar}$$

$$\dot{m}_1 = 50 \text{ kg/s}$$

$$z_1 = 0 \text{ m}$$

$$h_2 = 417.69 \text{ kJ/kg}$$

$$z_2 = 25 \text{ m}$$

$$\dot{Q}_{12} = -2 \text{ kW}$$

$$P_3 = 30 \text{ bar}$$

$$T_3 = 400^\circ\text{C}$$

$$P_4 = 5 \text{ bar}$$

$$T_4 = 180^\circ\text{C}$$

Find:

$$a.) \quad \dot{W}_{12} \text{ in kW}$$

$$b.) \quad \dot{m}_{34} \text{ in kg/s}$$

per Table A-3:

$$h_1 = 417.46 \frac{\text{kJ}}{\text{kg}}$$

$$\dots \left(\frac{\text{kg}}{\text{kg}} \right) \left(\frac{1 \text{ kJ}}{1000 \text{ J}} \right)$$

per Table A-4:

$$h_3 = 3230.9 \frac{\text{kJ}}{\text{kg}}$$

$$h_4 = 2812.0 \frac{\text{kJ}}{\text{kg}}$$

2 2 15 / 15

✓ - 0 pts Correct

- 15 pts Missing answer

- 1 pts Wrong calculation. $W = -25.8 \text{ kW}$

- 3 pts $h_1 = 417.46 \text{ kJ/kg}$. $W = -25.8 \text{ kW}$

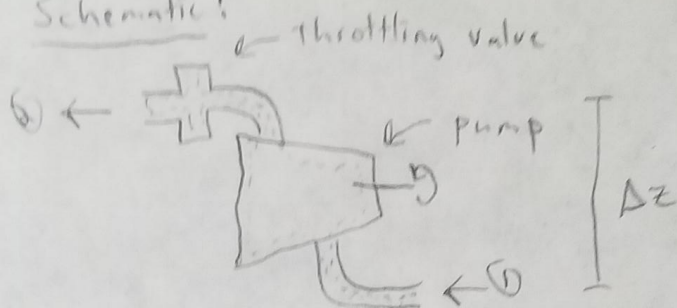
- 6 pts $(z_1 - z_2) = -25\text{m}$ (negative). $W = -25.8 \text{ kW}$

- 3 pts $h_3 = 3230.9 \text{ kJ/kg}$. Mass flow rate of steam = 0.0616 kg/s

3.)

Schematic:

(4.72)



(Given:

$$z_1 = 0$$

$$z_2 = 20 \text{ m}$$

$$\dot{V} = 150 \text{ m}^3/\text{s}$$

$$\dot{Q} = -0.6 \text{ MW}$$

$$g = 9.81 \text{ m/s}^2$$

Eng. Model:

Control Volume

Steady State

1 inlet / 1 outlet

$$\Delta T = 0 + \Delta p = 0 \Rightarrow \Delta h = 0$$

Electric Work

$$\Delta KE = 0$$

$$\frac{dm_{cv}}{dt} = 0 = \dot{m}_1 - \dot{m}_2$$

$$\dot{m}_1 = \dot{m}_2 = \dot{m}$$

$$\frac{dE_{cv}}{dt} = 0 = \dot{Q} - \dot{W} + \dot{m}((h_1 - h_2) + \frac{v_1^2 - v_2^2}{2} + (z_1 - z_2)g)$$

$$\dot{m} = \dot{V} \cdot \rho_{\text{water}} = 150 \frac{\text{m}^3}{\text{s}} \cdot 1 \frac{\text{g}}{\text{cm}^3} \cdot \left(\frac{100 \text{ cm}}{1 \text{ m}}\right)^3 \cdot \left(\frac{1 \text{ kg}}{1000 \text{ g}}\right)$$

$$\dot{m} = 1.50 \times 10^5 \text{ kg/s}$$

$$\dot{W} = -0.6 \text{ MW} + 1.50 \times 10^5 \frac{\text{kg}}{\text{s}} \left(9.81 \frac{\text{m}}{\text{s}^2} (-20 \text{ m}) \right) \cdot \frac{1 \text{ MW}}{10^6 \text{ W}}$$

$$\dot{W} = -30.03 \text{ MW}$$

The power generated will be less than the power required to pump because the heat lost by the system is still lost. $\dot{W}$ is expected to be positive, but $\dot{Q}$ stays negative.

33 20 / 20

- 0 pts Correct

✓ - 5 pts power requirement = -W

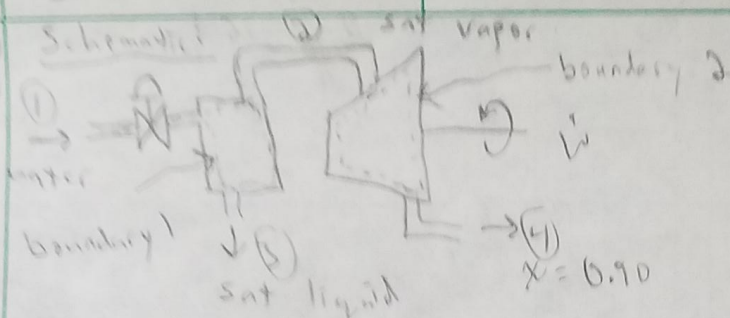
- 20 pts No submission

- 8 pts wrong calculation for W

✓ + 5 pts Extra points from Question 6

4)
(9.78)

Schematic



Eng. Model:

2 Control Volumes

Steady State

1 Inlet / 2 outlets and 1 inlet / 1 outlet

 $\Delta KE = \Delta PE = 0$ $\dot{Q} = 0$

$$\frac{dm_1}{dt} = \dot{m}_1 - \dot{m}_2 - \dot{m}_3 = 0 \quad \frac{dm_2}{dt} = \dot{m}_{ae} - \dot{m}_4 = 0$$

$$\dot{m}_1 = \dot{m}_2 + \dot{m}_3 \quad \dot{m}_2 = \dot{m}_4$$

$$\frac{dE_1}{dt} = \dot{m}_1 h_1 - \dot{m}_2 h_2 - \dot{m}_3 h_3 = 0$$

$$0 = \dot{m}_1 (h_1 - h_3) - \dot{m}_2 (h_2 - h_3)$$

$$\dot{m}_2 = \frac{\dot{m}_1 (h_1 - h_3)}{(h_2 - h_3)}$$

$$\dot{m}_2 = \frac{5 \text{ kg/s} (763.22 - 604.74)}{(2738.6 - 604.74)}$$

$$\dot{m}_2 = 0.3713 \frac{\text{kg}}{\text{s}}$$

$$h_4 = X(h_{4,g}) + (1-X)(h_{4,f})$$

$$h_4 = 0.9(2577.0 \frac{\text{kJ}}{\text{kg}}) + 0.1(173.88 \frac{\text{kJ}}{\text{kg}})$$

$$h_4 = 2336.7 \text{ kJ/kg}$$

$$\frac{dE_2}{dt} = -\dot{W} + \dot{m}_2 (h_2 - h_4) = 0$$

$$\dot{W} = 0.3713 \frac{\text{kg}}{\text{s}} \left(2738.6 \frac{\text{kJ}}{\text{kg}} - 2336.7 \frac{\text{kJ}}{\text{kg}} \right) \left(\frac{1 \text{ hp}}{0.7457 \text{ kW}} \right)$$

$$\dot{W} = 200.1 \text{ hp}$$

Given:

$$P_1 = 15 \text{ bar}$$

$$T_1 = 180^\circ \text{C}$$

$$\dot{m}_1 = 5 \text{ kg/s}$$

$$P_{2e} = 4 \text{ bar}$$

$$P_{3e} = 4 \text{ bar}$$

$$P_3 = 4 \text{ bar}$$

$$P_4 = 6.08 \text{ bar}$$

$$X_4 = 0.90$$

Find:

$$\dot{W} \text{ in hp}$$

From Table A-2:

$$h_1 = 763.22 \frac{\text{kJ}}{\text{kg}}$$

From Table A-3:

$$h_2 = 2738.6 \text{ kJ/kg}$$

$$h_3 = 604.74 \text{ kJ/kg}$$

From Table A-2:

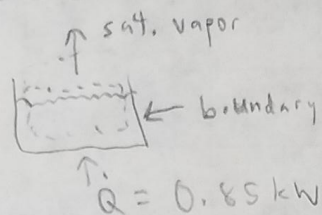
$$h_{4,f} = 173.88 \text{ kJ/kg}$$

$$h_{4,g} = 2577.0 \text{ kJ/kg}$$

4 4 15 / 15

✓ - 0 pts Correct

- 15 pts Missing answer
- 2 pts Did not define the control volumes
- 2 pts Did not state the engineering model
- 3 pts Should have used m_2 to calculate work. $W = 199.9$ hp
- 3 pts $h_1 \sim h_f(180 \text{ oC}) = 763.22$ kJ/kg. $m_2 = 0.371$ kg/s
- 1 pts $m_2 = ?$
- 1 pts Work is positive

5.
(4.87)Schematic:Eng. Model:

Control Volume
Unsteady State
1 outlet
 $\Delta PE = \Delta KE = 0$
 $\dot{W} = 0$

$$\frac{dm_{cv}}{dt} = -\dot{m}_e$$

State 1 is initial conditions
State 2 is saturated water
State 3 is saturated vapor

$$a.) U + KE + PE = Q - W$$

$$Q = U$$

$$\dot{Q} \Delta t = m(u_2 - u_1)$$

$$\Delta t = \frac{N_1 V_1 (u_2 - u_1)}{N_1 \cdot \dot{Q}}$$

$$\Delta t = \frac{(0.5 L) (10^{-3} m^3/L) (417.36 - 83.95) \frac{kJ}{kg}}{(1.0018 \times 10^{-3} \frac{m^3}{kg}) (0.85 \frac{kJ}{s})}$$

$$\Delta t = 195.8 s$$

$$b.) \Delta t = \frac{V_1 (u_3 - u_2)}{N_1 \cdot \dot{Q}}$$

$$\Delta t = \frac{(0.5 L) (10^{-3} m^3/L) (2506.1 - 417.36) \frac{kJ}{kg}}{(1.0018 \times 10^{-3} \frac{m^3}{kg}) (0.85 \frac{kJ}{s})}$$

$$\Delta t = 1226 s$$

Given:

$$V_1 = 0.5 L$$

$$T_1 = 20^\circ C$$

$$P_1 = 1 bar$$

$$\dot{Q} = 0.85 kW$$

$$P_2 = P_3 = 1 bar$$

Find:

a.) Time required
for water to
start boiling

b.) Time required
for all water to
evaporate after
boiling starts.

Per Table A-2:

$$N_1 = 1.0018 \times 10^3 \frac{m^3}{kg}$$

$$u_1 = 83.95 \frac{kJ}{kg}$$

Per Table A-3:

$$u_2 = 417.36 \frac{kJ}{kg}$$

$$u_3 = 2506.1 \frac{kJ}{kg}$$

5 5 15 / 15

- 0 pts Correct
- 15 pts Missing or wrong answer
- ✓ - 2 pts $W_{1-2} = 0.002 \text{ kJ}$
- 7.5 pts b) $t = 1328.3 \text{ s}$ (22.1 min)
- ✓ - 2 pts $W_{2-3} = -0.0522 \text{ kJ}$
- ✓ - 2 pts $h_e = 2675.5 \text{ kJ/kg}$
- 3 pts b) calculation is not clear.
- 13 pts Incomplete answer
- 2 pts $u_2 = 417.36 \text{ kJ/kg}$
- ✓ + 6 pts Extra points from Question 6

Fan

Blows air from the intake port to the exhaust port via an electrically worked impeller.



Resistive Coils

Electricity runs through these coils. The high resistivity of the coil material makes the coils heat up, thus heating the air blown through them.

Breaker

"Breaks" the electric circuit if the current suddenly spikes.



* Switch



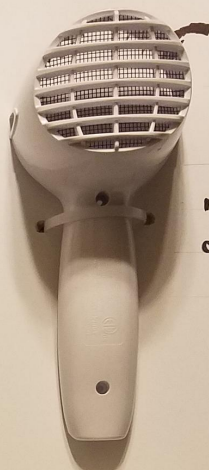
Heat Shield

Protects the exhaust frame from the heat generated by the coils.



Intake Port

This is where room temperature air comes in.



Exhaust Port

This is where heated air comes out.



Switch *

This controls the fan speed.

6 6 20 / 20

- 0 pts Correct

- 20 pts Missing answer

✓ + 20 pts Extra points for making the board

- 10 pts Missing or incomplete explanation of how it works and/or missing schematics of important parts (eg: nozzle, fan, switch, heating elements, cord etc).

- 5 pts Missing source for the drawing

- 2 pts Missing schematics of the air flow

- 10 pts Missing drawing

Wow! Great job!